ONLINE COURT RESERVATION SYSTEM

Updated Full Feature Specification

Purpose: A complete feature reference for an online court reservation platform covering dynamic booking, courts, members, matchmaking, waitlists, payments, memberships, administration, notifications, dashboards, reporting, and audit controls.

Recommended approach: Treat the booking engine as the core of the system and make business rules configurable rather than hard-coded.

1. Court Management

- Sport category management.
- Court creation and configuration.
- Court name/number and branch/location.
- Court capacity: configurable minimum and maximum players.
- Court status: Available, Maintenance, Closed, Temporarily Blocked.
- Opening and closing hours.
- Booking interval configuration.
- Minimum and maximum booking duration.
- Court-specific pricing where required.
- Court maintenance periods.
- Court utilization tracking.

2. Court Schedule & Calendar

- Daily and weekly court schedules.
- Calendar view for customers, staff, and administrators.
- Available, reserved, held, blocked, maintenance, and closed time slots.
- Recurring schedules.
- Special schedules and holidays.
- Schedule overrides for individual dates.
- Multi-court calendar for administrators.
- Date and time filtering.

3. Dynamic Booking Engine

- Add booking / create reservation.
- Online customer booking.
- Public booking where permitted.
- Select sport, court, date, time, duration, and players.
- Dynamic availability calculation.
- Conflict detection and double-booking prevention.
- Booking validation before confirmation.
- Configurable duration units; initial example: 30, 60, and 90 minutes.
- Configurable minimum and maximum capacity.
- Daily booking limits.
- Advance booking window.
- Membership-based booking restrictions.
- Price calculation.
- Booking confirmation.
- Booking history.

- Booking status tracking.

4. Booking Hold & Concurrency Protection

- Temporarily hold a selected court/time slot during checkout.
- Prevent another customer from claiming the same slot while payment is in progress.
- Configurable hold duration, such as 10 minutes.
- Release expired holds automatically.
- Convert a successful hold into a confirmed booking after successful payment.
- Record expired or failed checkout attempts.

5. Booking Status & Lifecycle

- DRAFT.
- PENDING_PAYMENT.
- CONFIRMED.
- CHECKED_IN.
- IN_PROGRESS.
- COMPLETED.
- CANCELLED.
- RESCHEDULED.
- NO_SHOW.
- EXPIRED.
- FAILED.
- Maintain status history for every booking.

6. Daily Limits & Booking Rules

- Maximum number of bookings per member per day.
- Maximum total booking minutes per day.
- Maximum future bookings.
- Advance booking window by membership tier.
- Court-specific restrictions.
- Sport-specific restrictions.
- Time-of-day restrictions.
- Special event or tournament restrictions.
- Configurable rules so administrators can change limits without code changes.

7. Cancellation & Refund Policies

- Configurable cancellation policy.
- Initial rule: cancellations within 24 hours receive no refund.
- Cancellation more than 24 hours before the booking can be eligible for refund.
- Optional partial-refund rules for different time windows.
- Distinguish customer cancellation, admin cancellation, court closure, and no-show.

- Record cancellation reason.
- Calculate eligible refund amount.
- Record refund status and transaction reference.

8. Rescheduling

- Customer rescheduling where permitted.
- Administrator rescheduling.
- Availability validation for the new date/time.
- Rescheduling cut-off policy.
- Price difference calculation where applicable.
- Additional payment or refund when the booking price changes.
- Preserve original booking history.
- Record reschedule reason.

9. Members & Players

- Member registration and profile.
- Membership status and expiration.
- Player management.
- Registered members as players.
- Guest players.
- Player count validation against court capacity.
- Booking history per member/player.
- Player invitations.
- Participant/session management.

10. Membership Management

- Membership types/levels.
- Membership start and expiration dates.
- Active, Pending, Expired, Suspended, and Cancelled states.
- Membership renewal.
- Membership benefits.
- Membership-specific pricing or discounts.
- Membership access rules.
- Membership expiration notifications.

11. Tiered Access & Privileges

- Configurable membership tiers such as Basic, Premium, and VIP.
- Priority booking.
- Different advance booking windows.
- Premium or restricted court access.
- Discounts.

- Special cancellation/rescheduling privileges.
- Tier-specific daily limits.
- Tier-specific duration or capacity rules.
- Administrator configuration of privileges.

12. Match Making / Find a Player

- Create a request to find additional players.
- Specify sport and preferred date/time.
- Specify required number of players.
- Optional skill-level matching.
- Optional location and availability matching.
- Send invitations to compatible players.
- Accept or decline invitation.
- Automatically update session/player list after acceptance.
- Close matchmaking when the session is full.

13. Waitlist Management

- Join waitlist when a court/session is full.
- Waitlist position.
- Automatic promotion when a slot becomes available.
- Notification when a slot becomes available.
- Configurable claim/response window.
- Automatically move to the next person when the claim expires.
- Waitlist cancellation.
- Waitlist history and audit trail.

14. Payments

- Online payment.
- E-wallet payment.
- Card or other supported payment methods.
- Cash/manual payment where required.
- Payment status: Pending, Paid, Failed, Refunded, Partially Refunded.
- Payment gateway reference.
- Payment confirmation.
- Failed-payment handling.
- Refund processing.
- Partial refund support.
- Payment reconciliation.

15. Transaction History

- Customer transaction history.

- Administrator transaction history.
- Transaction ID.
- Booking ID.
- Member.
- Amount.
- Payment method.
- Transaction date.
- Payment status.
- Refund amount and status.
- Gateway/reference number.

16. Outstanding Payments / Aged Fees

- Outstanding or unpaid transactions.
- Due date tracking.
- Aging buckets such as 0–7, 8–30, 31–60, 61–90, and 90+ days.
- Filter by member, booking, date, status, and age.
- Outstanding balance summary.
- Payment follow-up status.
- Exportable report.

17. Admin & Operational Controls

- Manual booking.
- Court blocking.
- Temporary court closure.
- Maintenance blocking.
- Tournament/event blocking.
- Limited booking windows.
- Manual cancellation.
- Manual rescheduling.
- Manual refund where authorized.
- Override selected booking rules with permission.
- Operational notes and reasons.
- Role-based permissions.

18. Notifications

- Booking confirmation.
- Booking reminder.
- Booking cancellation.
- Booking rescheduling.
- Payment success.

- Payment failure.
- Refund processed.
- Waitlist availability.
- Waitlist promotion.
- Claim expiration.
- Membership expiration.
- Operational/court closure notifications.
- In-app notifications, with email/SMS/push support where integrated.

19. Customer Dashboard

- Upcoming bookings.
- Today's bookings.
- Quick Book.
- Available courts.
- Sport category.
- Membership information.
- Payment balance.
- Notifications.
- Booking history.
- Transaction history.
- Search.
- Date and time filters.

20. Admin Dashboard

- Daily booking summary.
- Sport summary.
- Court utilization.
- Today's revenue.
- Confirmed, pending, cancelled, completed, and failed bookings.
- Pending/outstanding payments.
- Failed transactions.
- Waitlist activity.
- Membership statistics.
- Upcoming sessions.
- Court status.
- Search and filters.
- Date/time range reporting.

21. Failure & Exception Monitoring

- Failed booking tracking.

- Failed payment tracking.
- Payment gateway timeout tracking.
- Refund failure tracking.
- Notification failure tracking.
- Waitlist assignment failure tracking.
- Record failure stage, court, date/time, booking ID, member, and reason.
- Retry or resolution workflow for recoverable failures.

22. Reports

- Daily booking report.
- Court utilization report.
- Sport category report.
- Revenue report.
- Payment report.
- Outstanding/aged fees report.
- Cancellation report.
- Rescheduling report.
- No-show report.
- Membership report.
- Waitlist report.
- Failed transaction report.
- Booking activity/history report.
- Export to CSV/PDF where required.

23. Audit & History

- Booking status history.
- Cancellation history.
- Rescheduling history.
- Payment history.
- Refund history.
- Administrative changes.
- Court schedule changes.
- Court blocking history.
- Membership changes.
- Record who performed the action, when it occurred, and what changed.

24. Recommended Database Modules

Module	Suggested Tables
Users & Membership	users, members, membership_types, memberships
Sports & Courts	sports, courts, court_schedules, court_blocks
Booking	bookings, booking_players, booking_holds, booking_status_histories
Matchmaking & Waitlist	matchmaking_requests, waitlists
Payments	payments, payment_transactions, refunds
Notifications	notifications
Policies & Privileges	booking_policies, membership_privileges
Security & Audit	roles, permissions, audit_logs

25. Recommended Core Booking Relationship

Sport → Courts → Schedules/Blocks → Bookings → Players → Payment → Status History

Member → Membership → Bookings → Payments → Waitlists → Matchmaking

26. Recommended Customer Booking Workflow

- Login/Register → Select Sport → Select Date → View Available Courts → Select Court → Select Time → Select Duration → Add Players → System Validation → Booking Hold → Payment → Confirmation → Notification → My Bookings.
- If payment fails: record failure, release the hold after the configured timeout, and allow retry where appropriate.
- If the slot is unavailable: show alternatives or allow the customer to join a waitlist.

27. Recommended Admin Workflow

- Admin Dashboard → Courts/Schedules → Bookings → Members → Memberships → Waitlists → Matchmaking → Payments/Refunds → Reports → Notifications → Audit Logs.
- Administrators should be able to block courts, create manual bookings, cancel/reschedule authorized bookings, and review failed transactions according to role permissions.

28. Key Business Rules to Finalize Before Development

- Sports and number of courts.
- Capacity per court/sport.
- Allowed duration values and interval.
- Daily booking limit.
- Maximum future bookings.
- Advance booking window per membership tier.
- Cancellation and refund windows.
- Rescheduling cut-off and fees.
- Payment methods and payment gateway.
- Booking hold duration.
- Waitlist claim/expiration period.

- Matchmaking rules.
- Membership tiers and privileges.
- Court opening/closing schedules.
- Rules for court closures after an existing reservation.
- Admin roles and authorization levels.

Implementation recommendation: Finalize these business rules first, then create the ERD/database schema, booking state machine, API endpoints, and frontend screens. This reduces the risk of changing the booking engine after the UI has already been built.